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Test Certificate

Product Code:	RLD11DFB024BK
Description:	24f, Non-Metallic Flat FRP armoured Axial tube, GC/PE/NY/FRP/PE
Cable #:	40539235
Date Cert created:	March 2024
Design Number:	23-21
Product Family:	RLx1xxFB^{+++}XX

TEST PARAMETER / METHOD	AFL ACCEPTANCE CRITERIA	ACHIEVED	RESULT*
Tensile Strength IEC 60794-1-21 (E1)	**6**kN 30min, < 0.6% fibre strain, No att. Increase after test, < 0.1dB during test, @ 1550nm	**6**kN <0.6% fs $\Delta\alpha \leq 0.01$dB No att. Increase after test	**Pass**
Bend under tension IEC 60794-1-21 (E18A)	r_b = **20** x OD 360°, 1 turn, 1 min No fibre break, no damage to sheath or core structure and No attn. change throughout test	r_b= **20** x OD$_{measured}$, No breaks or damage and no attenuation change	**Pass**
Crush (short term) IEC 60794-1-21 (E3)	**4**kN/100mm Duration: 10min 3 positions, 1 @ reversal. No damage to sheath or core structure and No att. Increase after test	**4kN**/100mm No damage and no attenuation increase after test	**Pass**
Crush (long term) IEC 60794-1-21 (E3)	**3**kN/100mm Duration: 120min 3 positions, 1 @ reversal. No damage to sheath or core structure and No att. Increase during and after test	**3kN**/100mm No damage and no attenuation increase during and after test	**Pass**
Impact IEC 60794-1-21 (E4)	Weight: **1.5**kg Height: **1**m, 300mm rad anvil, 3 impacts, After 5 mins No fibre break, no damage to sheath or core structure and No attn. change throughout test	Weight: **1.5**kg Height: **1**m, No breaks or damage and no attenuation change	**Pass**
Repeated Bending IEC 60794-1-21 (E6)	rb = 20 x OD 360° loop, 1 turn, 25 cycle No att. Change after test	rb = ~18 x OD (400mm Dia.) No attenuation change	**Pass**
Torsion IEC 60794-1-21-(E7)	Sample length: 1 m Rotation: a) 180º clockwise, b) return to starting position, c) 180º anticlockwise, d) return to starting position. Four movements = one cycle. Complete 10 cycles (a to d) in one minute maximum During the final tenth cycle at a), c) and after completion (no rotation) check transmitting fibres. No fibre breaks, no damage to the sheath or to the core structure and no attenuation change throughout test	No breaks or damage and no attenuation change	**Pass**
Bend IEC 60794-1-21 (E11)	r_b = **10** x OD 360° loop, 4 turn, 3 cycles No att. Change throughout test	r_b= **10** x OD$_{measured}$, No breaks or damage and no attenuation change	**Pass**
Temperature Cycle IEC 60794-1-22 (F1)	Temperature Range: **-40°C to +70°C** @ 1550nm $\Delta\alpha_{max\ Last\ cycle\ TA2\ and\ TB2} \leq 0.15$dB $\Delta\alpha_{max\ TA1\ (-30°C)\ and\ TB1\ (+60°C)} < 0.05$dB $\Delta\alpha_{max\ completion\ (ambient)} < 0.05$dB	Temperature Range: **-40°C to +70°C** @ 1550nm $\Delta\alpha_{max\ Last\ cycle\ TA2\ and\ TB2} \leq 0.10$dB $\Delta\alpha_{max\ TA1\ (-30°C)\ and\ TB1\ (+60°C)} < 0.05$dB $\Delta\alpha_{max\ completion\ (ambient)} < 0.05$dB	**Pass**
Water Penetration IEC 60794-1-22 (F5B)	Sample length: 3m Water Height: 1m No water after 24 hours	No Water detected	**Pass**
Tested By: Karlo Lucic	**Verified By: Shashini Subasinghe**	**Approved By: Jim Boukouvalas**	